04 Networking Basics and Ports

1. Introduction to Networking

Computer networking is the process of connecting two or more devices so they can communicate and share data, files, applications and resources.

Example: When you open a website, your computer communicates with a web server through a network.

Networking knowledge is important in cybersecurity because it helps us understand how data travels, how devices communicate and how attackers may misuse network services.

2. What Is an IP Address?

An **IP (Internet Protocol) address** is a logical address used to identify a network interface and help route data between devices across networks.

Example: 192.168.1.10

Types of IP Addresses

There are two main versions of IP addresses:

- IPv4
- IPv6

IPv4 vs IPv6

Basis	**IPv4**	**IPv6**
Full form	Internet Protocol version 4	Internet Protocol version 6
Address size	32 bits	128 bits
Format	Decimal numbers separated by dots	Hexadecimal groups separated by colons
Example	192.168.1.10	2001:db8::10
Address space	About 4.3 billion addresses	Extremely large address space
Representation	Four decimal octets	Up to eight hexadecimal groups
Broadcast	Supports broadcast	Does not use broadcast; uses multicast and other mechanisms
Main purpose	Identifies devices and routes packets	Provides a much larger address space and modern IP capabilities

Remember: IPv4 uses 32 bits; IPv6 uses 128 bits.

3. What Is a MAC Address?

A **MAC (Media Access Control) address** is a link-layer address used to identify a network interface for communication on a local network.

Example: 00:1A:2B:3C:4D:5E

Important Points

- A typical Ethernet or Wi-Fi MAC address is 48 bits long.
- It is commonly written in hexadecimal.
- It helps deliver frames within a local network.
- MAC addresses can sometimes be changed or spoofed.
- A MAC address is different from an IP address.

IP Address vs MAC Address

Basis	IP Address	MAC Address
Purpose	Logical addressing and routing	Local link-layer delivery
Layer	OSI Layer 3	OSI Layer 2
Example	192.168.1.10	00:1A:2B:3C:4D:5E
Assignment	Configured manually or dynamically	Usually assigned to an interface; can be changed
Scope	Can be used across networks	Normally used on the local network link

Example: When a computer sends data to another device on the same Ethernet network, it uses the destination IP address and the appropriate destination MAC address for local delivery.

4. What Is DNS?

DNS stands for Domain Name System.

DNS translates domain names into IP addresses and provides other naming information.

People remember names more easily than numerical IP addresses.

Example:

www.example.com → an IP address returned by DNS.

How DNS Works

1. You type a domain name into your browser.
2. Your computer checks available DNS information, including its cache.
3. If necessary, it queries a DNS resolver.
4. The resolver obtains the required DNS information.
5. Your computer uses the resulting IP address to connect to the destination.

Common DNS Record Types

Record	Purpose
A	Maps a hostname to an IPv4 address
AAAA	Maps a hostname to an IPv6 address
MX	Identifies mail servers for a domain
CNAME	Creates an alias for another hostname
TXT	Stores text information, often used for verification and email security
NS	Identifies authoritative name servers

Security Risks of DNS

- DNS spoofing or cache poisoning.
- DNS tunneling used to hide unauthorized communications.
- DNS misconfiguration.
- DNS-based phishing and redirection.

Protection: Use trusted DNS resolvers, keep DNS software updated and apply DNS security controls where appropriate.

5. What Is DHCP?

DHCP stands for Dynamic Host Configuration Protocol.

DHCP automatically provides network configuration to devices, such as:

- IP address.
- Subnet mask or prefix.
- Default gateway.
- DNS server addresses.

Without DHCP, network settings may need to be configured manually.

How DHCP Works: DORA

DORA represents the four traditional DHCPv4 address-assignment steps.

Step	Name	Meaning
1	Discover	The client searches for a DHCP server
2	Offer	The server offers an IP configuration
3	Request	The client requests the offered configuration
4	Acknowledge	The server confirms the lease

Memory trick: DORA = Discover, Offer, Request, Acknowledge.

DHCP Security Risks

- Rogue DHCP servers.
- DHCP starvation attacks.
- Incorrect network configuration.

Protection: Use DHCP snooping on supported switches, trusted ports and suitable network access controls.

6. What Is a Port?

A **network port** is a number used by transport protocols such as TCP and UDP to identify a service or application endpoint on a device.

An IP address identifies the destination network interface, while the port helps direct traffic to the correct service.

Example:

192.168.1.10:443

Here:

- IP address: 192.168.1.10
- Port: 443

Port Number Ranges

Range	Category	Description
0–1023	Well-known ports	Common standard services
1024–49151	Registered ports	Registered applications and services
49152–65535	Dynamic/private ports	Commonly used for temporary client connections

Port numbers are separate for TCP and UDP. The same numeric port can be used by both protocols for different services.

7. TCP vs UDP

TCP stands for Transmission Control Protocol.

TCP establishes a connection and provides reliable, ordered delivery of data. It retransmits missing data when needed.

UDP stands for User Datagram Protocol.

UDP sends datagrams without establishing a TCP-style connection. It does not itself guarantee delivery or ordering.

Basis	TCP	UDP
Connection	Connection-oriented	Connectionless
Reliability	Reliable delivery mechanisms	No built-in delivery guarantee
Ordering	Maintains ordered byte-stream delivery	No built-in ordering guarantee
Overhead	Generally higher	Generally lower
Examples	HTTPS over TCP, SSH, email	DNS queries commonly use UDP; voice and streaming applications may use UDP
Common risk	SYN flood attacks	UDP flood attacks

TCP Three-Way Handshake

TCP normally establishes a connection using three steps:

1. **SYN:** Client requests a connection.
2. **SYN-ACK:** Server acknowledges and responds.
3. **ACK:** Client acknowledges the response.

This establishes the TCP connection before ordinary application data is exchanged.

TCP vs UDP Memory Trick

- TCP = reliable, ordered delivery.
- UDP = lower overhead, no built-in delivery guarantee.

8. Private IP vs Public IP

Private IP Address

A private IP address is used inside private networks and is not directly routable across the public internet.

IPv4 private address ranges are:

- 10.0.0.0 – 10.255.255.255
- 172.16.0.0 – 172.31.255.255
- 192.168.0.0 – 192.168.255.255

Example: 192.168.1.20

Private addresses are commonly used in homes, offices and virtual machine labs.

Public IP Address

A public IP address is globally routable on the internet, subject to network routing and access controls.

Example: A publicly assigned IP address used by a website.

Comparison

Basis	**Private IP**	**Public IP**
Use	Internal networks	Internet-facing communication
Uniqueness	May be reused in separate private networks	Must be globally unique when assigned for public routing
Internet routing	Not directly routed on the public internet	Globally routable when properly advertised
Example	192.168.1.20	A globally assigned IPv4 address
Typical security	Protected through network boundaries and controls	Requires appropriate exposure and access controls

Important: A private IP does not automatically make a device secure. A public IP does not automatically mean a device is vulnerable. Security depends on configuration and controls.

9. What Is a Default Gateway?

A **default gateway** is the router or Layer 3 device that forwards traffic from a local network to destinations outside the local subnet when no more specific route applies.

Example:

- Computer IP: 192.168.1.20
- Subnet mask: 255.255.255.0
- Default gateway: 192.168.1.1

If the computer needs to contact a destination outside 192.168.1.0/24, it normally sends the traffic to the default gateway.

10. What Is a Subnet Mask?

A **subnet mask** identifies which bits of an IPv4 address belong to the network portion and which belong to the host portion.

Example:

- IP address: 192.168.1.20
- Subnet mask: 255.255.255.0
- CIDR notation: /24
- Network address: 192.168.1.0

In this example, the first 24 bits represent the network prefix.

Why Is a Subnet Mask Important?

- Identifies the local network.
- Helps a device determine whether a destination is local or remote.
- Supports network division and IP address planning.
- Helps configure routing and security boundaries.

Gateway vs Subnet Mask

Gateway	Subnet Mask
Forwards traffic to other networks	Defines the IPv4 network prefix
Usually an IP address of a router or Layer 3 interface	A bitmask, such as 255.255.255.0
Example: 192.168.1.1	Example: 255.255.255.0

11. Common Network Ports and Security Risks

A port being open does not automatically mean a system is vulnerable. Risk depends on the service, software version, configuration, authentication and network exposure.

Port	Protocol	Service	Purpose	Typical Security Risk
21	TCP	FTP control	Controls file transfers	Plaintext credentials and insecure configuration
22	TCP	SSH	Secure remote login and administration	Brute-force attempts, stolen keys or weak authentication
23	TCP	Telnet	Remote terminal access	Credentials and data may travel in plaintext
25	TCP	SMTP	Transfers email between mail servers	Open relay abuse, spam and misconfiguration
53	UDP/TCP	DNS	Domain name resolution and DNS operations	Spoofing, amplification abuse or misconfiguration
80	TCP	HTTP	Unencrypted web traffic	Data exposure if sensitive information is sent without protection
110	TCP	POP3	Retrieves email	Credentials and email may be exposed without TLS
143	TCP	IMAP	Accesses and manages email on a server	Credentials and email may be exposed without TLS
443	TCP	HTTPS	Encrypted web traffic over TLS	Weak TLS settings, vulnerable applications or stolen credentials
445	TCP	SMB	Windows file and printer sharing	Unauthorized file access, ransomware spread or exploitation of vulnerable SMB services
3306	TCP	MySQL	Database connections	Unauthorized database access or data exposure
3389	TCP/UDP	RDP	Remote Windows desktop access	Brute-force attempts, stolen credentials or exposed remote access

Important Port Security Recommendations

- Disable services that are not needed.
- Restrict administrative services such as SSH and RDP to trusted networks or VPN access.
- Avoid exposing databases such as MySQL directly to the internet.
- Prefer SFTP or another secure file transfer method instead of plaintext FTP.
- Use TLS for email protocols where supported and configured.
- Keep services and operating systems updated.
- Apply firewalls, strong authentication and logging.
- Review exposed ports regularly.

12. Basic Networking Commands

Command	Purpose	Example
ip a	Displays network interfaces and IP addresses on Linux	ip a
ip route	Displays the routing table and default gateway	ip route
ping	Tests basic IP reachability	ping 8.8.8.8
traceroute	Shows the network path toward a destination	traceroute example.com
nslookup	Queries DNS information	nslookup example.com
dig	Performs detailed DNS queries	dig example.com
ss -tulpn	Lists listening sockets and related process information when permitted	sudo ss -tulpn
ip neigh	Displays the local neighbor table	ip neigh

Note: Some Linux distributions require installing traceroute or dig. Network firewalls may also block ping or path-discovery traffic.

13. Practical Example

Suppose your Ubuntu VM has these settings:

- IP address: 192.168.56.10
- Subnet mask: 255.255.255.0
- Default gateway: 192.168.56.1
- DNS server: 192.168.56.1

When you open a website:

1. DNS resolves the domain name to an IP address.
2. Your computer checks its routing table.
3. If the destination is outside the local subnet, traffic is sent toward the default gateway.
4. TCP or UDP carries the application data, depending on the protocol.
5. The destination port identifies the appropriate service.
6. Firewalls and other security controls determine whether traffic is allowed.

These example values are illustrative. Your actual VM addresses depend on your network configuration.

14. Quick Revision Questions

Q1. What is an IP address?

An IP address is a logical address used to identify a network interface and route packets.

Q2. What is the difference between IPv4 and IPv6?

IPv4 uses 32-bit addresses, while IPv6 uses 128-bit addresses.

Q3. What is a MAC address?

A link-layer address used for local network communication.

Q4. What is DNS?

DNS translates domain names into IP addresses and provides other naming information.

Q5. What is DHCP?

DHCP automatically supplies IP configuration to network devices.

Q6. What is a port?

A number used by TCP or UDP to identify a service or application endpoint.

Q7. What is the difference between TCP and UDP?

TCP provides reliable, ordered delivery; UDP does not provide built-in delivery or ordering guarantees.

Q8. What is a default gateway?

The router or Layer 3 device used to reach destinations outside the local subnet when no more specific route applies.

Q9. What is a subnet mask?

It identifies the network prefix and host portion of an IPv4 address.

Q10. Why is port 445 important in cybersecurity?

It is commonly used by SMB for Windows file sharing and can create significant risk if exposed or improperly secured.

Q11. Which port is used by HTTPS?

TCP port 443 is the standard port for HTTPS over TCP.

Q12. Which port is commonly used by SSH?

TCP port 22.

Conclusion

Networking basics are essential for cybersecurity. Understanding IP and MAC addresses, DNS, DHCP, ports, TCP and UDP, gateways and subnet masks helps security professionals investigate network problems and recognize possible attacks. Learning the common ports also helps identify exposed services and apply appropriate security controls.